AADHRITA HACK24

EcoErase – Simple, Secure & Auditable Wiping Tool

By TECHCRAFTERS

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PROBLEM STATEMENT

India generates 1.75M+ tonnes of e-waste yearly, with devices hoarded over data breach fears. Costly, complex, unverifiable tools leave ₹50,000+ crore IT assets idle, slowing circular economy growth. A simple, secure, auditable solution is needed.

Problem

Create a cross-platform wiping tool (Windows, Linux, Android) that:

- Erases all data, including hidden areas.
- Issues tamper-proof digital certificates.
- Offers one-click UI with offline ISO/USB.
- Ensures NIST SP 800-88 compliance & third-party verification.

Impact

Boosts recycling trust, reduces hoarding, enables safe disposal, and supports India's circular economy.

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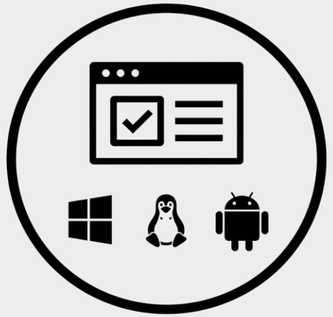
FEASIBILITY STUDY



Technical Feasibility:

- Supports multiple platforms (Windows, Linux, Android).
- Compliant with global standards (NIST SP 800-88).

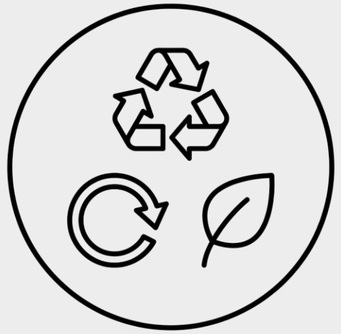
Operational Feasibility:

- One-click, user-friendly interface.
- Works both online & offline (bootable ISO/USB).

EcoErase

Economical Feasibility:

- Unlocks value from hoarded IT assets.
- Low-cost solution with high ROI.



Environmental Feasibility:

- Reduces e-waste by enabling reuse.
- Strengthens circular economy efforts.

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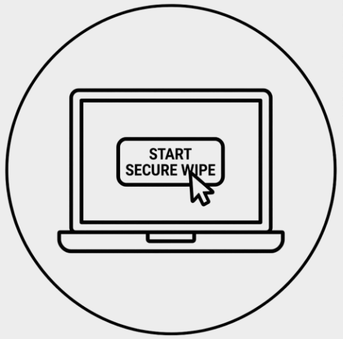
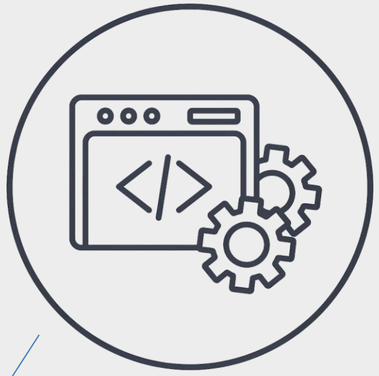
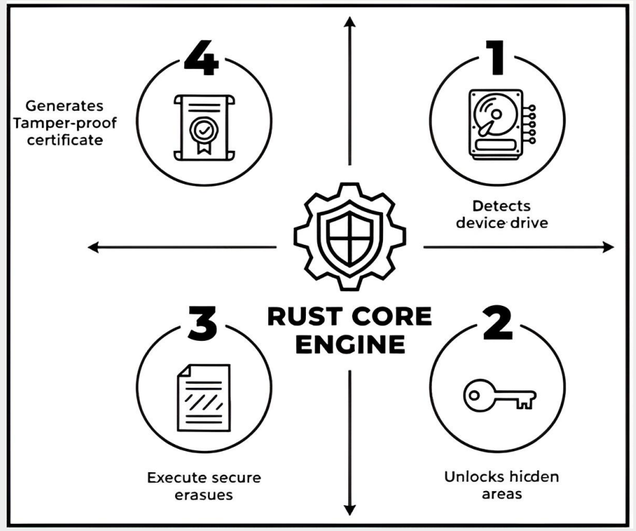
SOLUTION



(An intuitive one-click interface suitable for general public use)



(Calls the Rust Core Engine through Foreign Functional Interface (FFI) and Engine starts wiping)



(Generates Tamper-proof certificate after wiping successfully)

"Verifies from hash values generated before and after wipe"

"or QR code confirmation"

(So that no one can forge it)

(Third party verification)

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TECH STACK

1.Core Engine :

Rust : Memory-safe + high-performance language for erasure engine

2.Wrappers / OS Integration :

Windows: C# calls DLL via P/Invoke (Platform Invocation Services).
Linux: GTK/Qt/Python calls .so via FFI (Foreign Function Interface).
Android: Kotlin/Java calls .so via JNI (Java Native Interface).

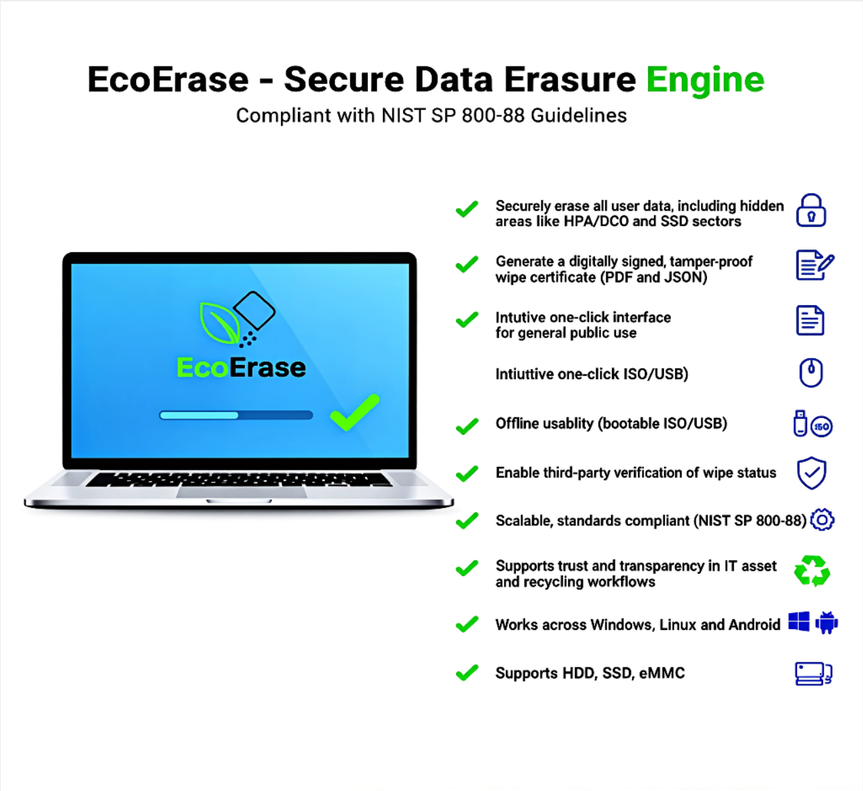
3.Frontend (User Interface):

Electron / Tauri (with React) → Unified cross-platform GUI.

4. Backend / Certificate Handling:

Node.js or Python (Flask/Fast_API) → For certificate generation logic.

OpenSSL / Rust_Crypto → For digital signatures & encryption.



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IMPACT AND BENEFITS

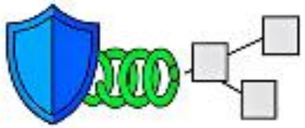
- Enterprises: Secure, verifiable bulk disposal.

- Individuals: Safe reuse/resale without the fear of data theft.

- Environment: Less e-waste more recycling.

- Advanced cryptographic erasures used for data irrecoverability.

- Low-cost software solution.



- Robust, Flexible, Scalable.

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